# IDA LAB-2

## This lab consists of FIVE Practical

## Practical-1:

- Demonstrate Binomial distribution using R

**Binomial Distribution:**

The binomial distribution is a discrete probability distribution. It describes the outcome of n independent trials in an experiment. Each trial is assumed to have only two outcomes, either success or failure. If the probability of a successful trial is p, then the probability of having x successful outcomes in an experiment of n independent trials is as follows.

$$f(x) = \binom{n}{x} p^x (1-p)^{(n-x)} \quad where\ x = 0, 1, 2, \ldots, n$$

**Problem**

- Suppose there are twelve multiple choice questions in an English class quiz. Each question has five possible answers, and only one of them is correct. Find the probability of having four or less correct answers if a student attempts to answer every question at random.

**Solution**

- Since only one out of five possible answers is correct, the probability of answering a question correctly by random is 1/5=0.2. We can find the probability of having exactly 4 correct answers by random attempts as follows.

```
> dbinom(4, size=12, prob=0.2)
[1] 0.1329
```

- To find the probability of having four or less correct answers by random attempts, we apply the function dbinom with x = 0,…,4.

```
> dbinom(0, size=12, prob=0.2) +
+ dbinom(1, size=12, prob=0.2) +
+ dbinom(2, size=12, prob=0.2) +
+ dbinom(3, size=12, prob=0.2) +
+ dbinom(4, size=12, prob=0.2)
[1] 0.9274
```

- Alternatively, we can use the cumulative probability function for binomial distribution pbinom.

```
> pbinom(4, size=12, prob=0.2)
[1] 0.92744
```

**Answer**

- The probability of four or less questions answered correctly by random in a twelve-question multiple choice quiz is 92.7%.

# Practical-2:

- Demonstrate Poisson distribution using R

**Poisson Distribution:**

The Poisson distribution is the probability distribution of independent event occurrences in an interval. If λ is the mean occurrence per interval, then the probability of having x occurrences within a given interval is:

$$f(x) = \frac{\lambda^x e^{-\lambda}}{x!} \quad where\ x = 0, 1, 2, 3, \ldots$$

**Problem**

- If there are twelve cars crossing a bridge per minute on average, find the probability of having seventeen or more cars crossing the bridge in a particular minute.

**Solution**

- The probability of having sixteen or less cars crossing the bridge in a particular minute is given by the function ppois.

```
> ppois(16, lambda=12)   # lower tail
[1] 0.89871
```

- Hence the probability of having seventeen or more cars crossing the bridge in a minute is in the upper tail of the probability density function.

```
> ppois(16, lambda=12, lower=FALSE)   # upper tail
[1] 0.10129
```

**Answer**

- If there are twelve cars crossing a bridge per minute on average, the probability of having seventeen or more cars crossing the bridge in a particular minute is 10.1%.

# Practical-3:

- Demonstrate Normal distribution using R

**Normal Distribution:**

The normal distribution is defined by the following probability density function, where μ is the population mean and $\sigma^2$ is the variance.

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-(x-\mu)^2/2\sigma^2}$$

If a random variable X follows the normal distribution, then we write:

$$X \sim N(\mu, \sigma^2)$$

In particular, the normal distribution with $\mu = 0$ and $\sigma = 1$ is called the standard normal distribution, and is denoted as N(0,1).

The normal distribution is important because of the Central Limit Theorem, which states that the population of all possible samples of size n from a population with mean $\mu$ and variance $\sigma^2$ approaches a normal distribution with mean $\mu$ and $\sigma^2/n$ when n approaches infinity.

**Problem**

- Assume that the test scores of a college entrance exam fits a normal distribution. Furthermore, the mean test score is 72, and the standard deviation is 15.2. What is the percentage of students scoring 84 or more in the exam?

**Solution**

- We apply the function pnorm of the normal distribution with mean 72 and standard deviation 15.2. Since we are looking for the percentage of students scoring higher than 84, we are interested in the upper tail of the normal distribution.

```
> pnorm(84, mean=72, sd=15.2, lower.tail=FALSE)
[1] 0.21492
```

**Answer**

- The percentage of students scoring 84 or more in the college entrance exam is 21.5%.

## Practical-4:

- Implement and perform the following:

**(1).** Probability that a normal random variable with mean 22 and variance 25
*(i)* lies between 16.2 and 27.5
*(ii)* is greater than 29
*(iii)* is less than 17
*(iv)* is less than 15 or greater than 25

**(2).** Probability that in 60 tosses of a fair coin the head comes up
*(i)* 20,25 or 30 times
*(ii)* less than 20 times
*(iii)* between 20 and 30 times

**(3).** A random variable X has Poisson distribution with mean 7. Find the probability that
*(i)* X is less than 5
*(ii)* X is greater than 10
*(iii)* X is between 4 and 16

**(4).** Suppose that there are 10 independent trials, and that the probability of success on each trial is 0.6, find the probability of 5 successes.

**(5).** At Kerr Pharmacy owner determined that there is a 0.4 chance of any one employee being late. 5 employees are in the pharmacy. find the probabilities of 0,1,2,3,4, or 5 employees being late simultaneously?

**(6).** for binomial distribution with n=10 and p=0.45, find:
(a) P(r=8)
(b) P(r>4)
(c) P(r<=6)

**(7).** Suppose toy weights produced at LEGO Toys Works have weights that are normally distributed with mean 17.46 grams and variance 375.67 grams. What is the probability that a randomly chosen toy weighs more than 19 grams?

## Practical-5:

- Implement and perform the following:

- Write a function that will simulate numbers from a given distribution. The function should take one argument for the number of samples and a second argument which specifies the distribution (Normal, Poisson or Binomial). The function should be able to handle additional parameters depending on the distribution chosen, e.g. a 'lambda' for Poisson samples.